# LCEC-IO — EtherCAT slave I/O board

Root sheet, milestone M5: Power (2), MCU (3), EtherCAT (4), Inputs (5), MPG (6), Relays (7), Analog (8). The block diagram, mounting holes and PE terminal follow in M6.

Power

File: 02_power.kicad_sch

MPG

MPG_A
MPG_B
RELAY_OE
OE_FIELD

File: 06_mpg.kicad_sch

Inputs

DI[1..8]
SEL[1..12]

File: 05_inputs.kicad_sch

MCU

MPG_A
MPG_B
RELAY_OE
OE_FIELD
DI[1..8]
SEL[1..12]
CLK_ESC
ESC_CS
ESC_SCK
ESC_MOSI
ESC_RST
ESC_MISO
ESC_IRQ
ESC_SYNC0
ESC_SYNC1
ERR_LED
DO[1..8]
OE_FIELD
DAC_SCK
DAC_MOSI
DAC_CS
AO_CLAMP

File: 03_mcu.kicad_sch

EtherCAT

CLK_ESC
ESC_CS
ESC_SCK
ESC_MOSI
ESC_RST
ESC_MISO
ESC_IRQ
ESC_SYNC0
ESC_SYNC1
ERR_LED

File: 04_ethercat.kicad_sch

Relays

DO[1..8]
OE_FIELD

File: 07_relays.kicad_sch

Analog

DAC_SCK
DAC_MOSI
DAC_CS
AO_CLAMP

File: 08_analog.kicad_sch

Root sheet: all eight sheets; the block diagram, mounting holes and PE follow in M6

**Yantra Innovation — LCEC-IO**

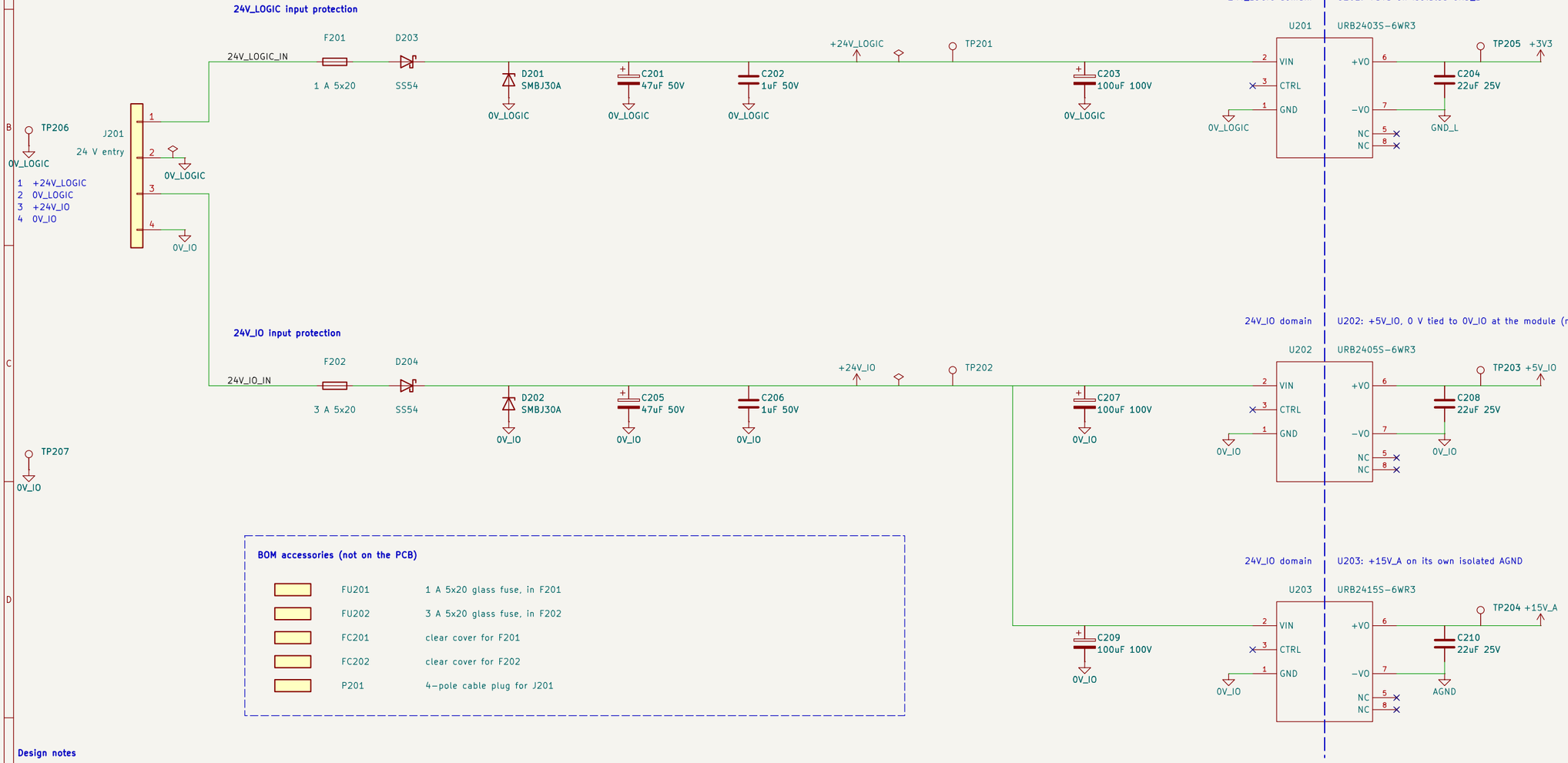
Sheet: /
File: lcec-io.kicad_sch

***Title: LCEC-IO — EtherCAT slave I/O board***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 1/13 |

# 2 POWER – 24 V entries, protection, three isolated DC–DC modules

Input 24 VDC +/–20 % = 19.2 … 28.8 V. Per input: fuse > SS54 reverse–polarity diode > SMBJ30A TVS > 47 uF + 1 uF, then a 100 uF/100 V input capacitor at each URB module (URB datasheet Cin).




**BOM accessories (not on the PCB)**

| | | |
|---|---|---|
| FU201 | 1 A 5x20 glass fuse, in F201 |
| FU202 | 3 A 5x20 glass fuse, in F202 |
| FC201 | clear cover for F201 |
| FC202 | clear cover for F202 |
| P201 | 4–pole cable plug for J201 |

**Design notes**

Loads: +3V3 about 0.4 A of the module's 1.35 A (30 %). 24V_LOGIC input about 0.3 A at full module load: 1 A fuse.

24V_IO: 5 V module 7.3 W + 15 V module 7 W at full load = 0.75 A, plus relay coils 0.21 A and sensors through the PTCs 1 A: 3 A fuse.

Input inrush I2t into the input capacitors stays below 1.1 A2s; the fuses are rated 4.26 A2s (1 A) and 12.5 A2s (3 A).

URB input 9–36 V (40 V continuous, 50 V for 1 s). SMBJ30A: 30 V stand–off above the 28.8 V maximum input, clamps 48.4 V at 12.4 A.

CTRL left open = module on. NC pins are not connected inside the module.

Two 24 V entries, protection, three isolated DC–DC modules

**Yantra Innovation – LCEC–IO**

Sheet: /Power/

File: 02_power.kicad_sch

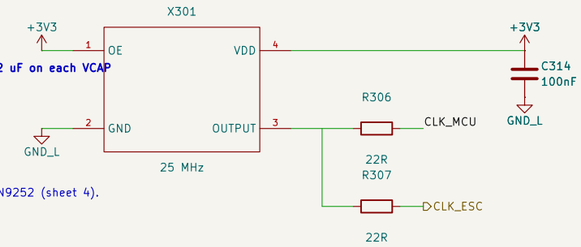
***Title: LCEC–IO – 2 Power***

| | | |
|---|---|---|
| Size: A3 | Date: 2026–11–06 | **Rev: A** |
| KiCad E.D.A. 10.0.8 | | Id: 2/13 |

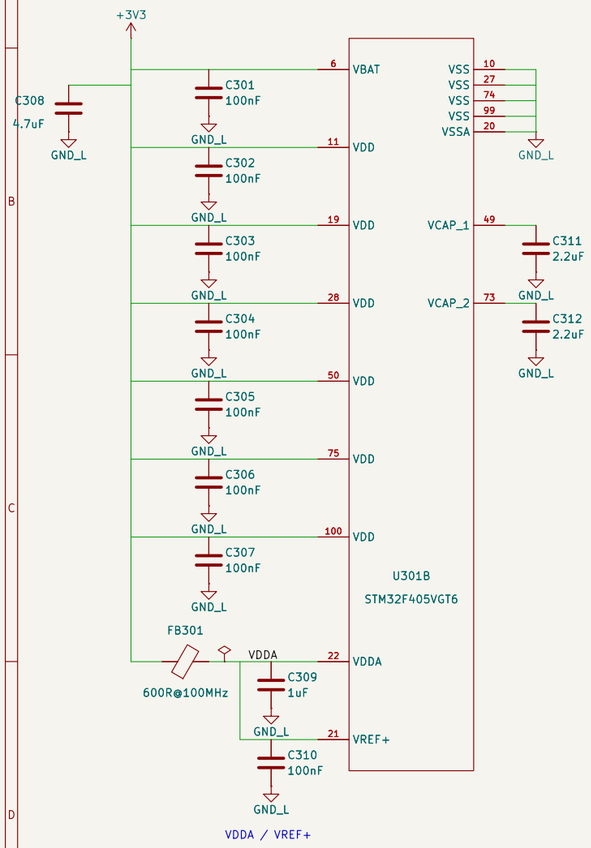
# 3 MCU — STM32F405VGT6, shared 25 MHz clock, SWD, UART2, heartbeat LED

51 signals used (pin map in docs/plan.md), PH1 is a test pad, the 30 pins of ports D and E are free for future use (no-connect flags). Firmware: HSE bypass, PLLM 25 / PLLN 336 / PLLP 2 = 168 MHz, IWDG enabled in option bytes.

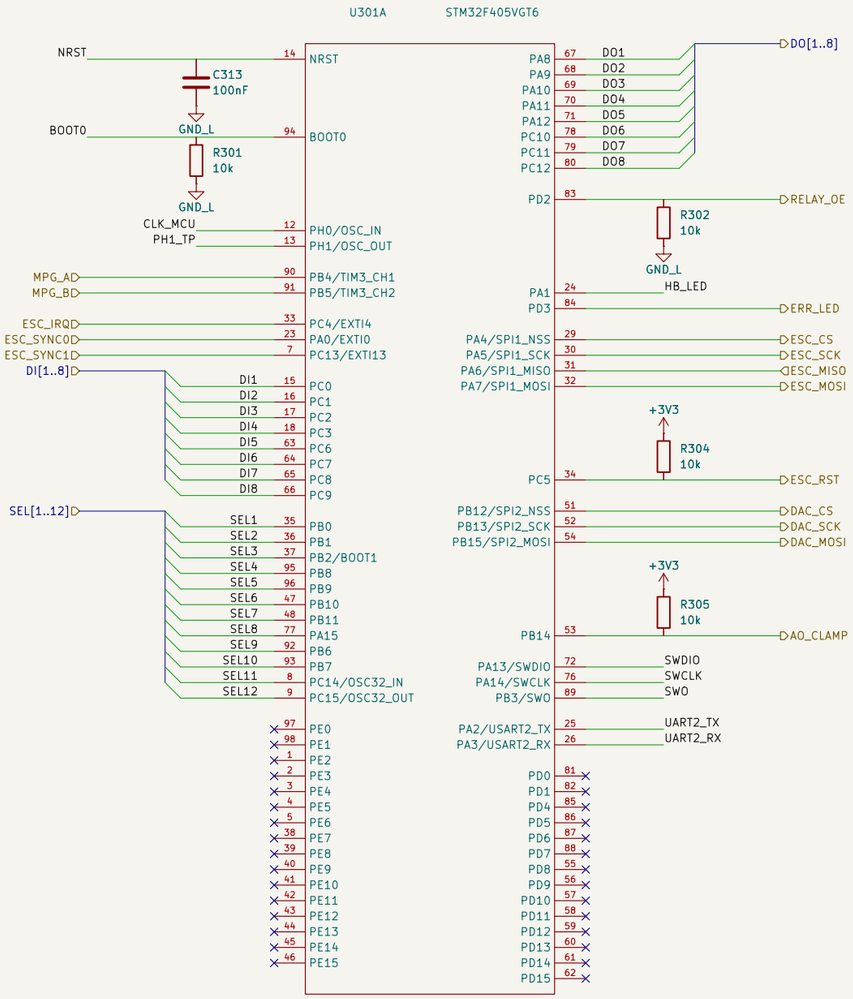
**U301 power: 100 nF beside each VDD pin and VBAT, 4.7 uF bulk, VDDA through a ferrite, 2.2 uF on each VCAP**



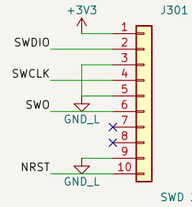

OE tied high = oscillator running. 22 R series resistors to PH0 (HSE bypass) and to the LAN9252 (sheet 4).



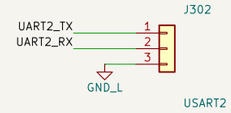

VDDA / VREF+




SWD (Cortex 10-pin): 1 VTref, 2 SWDIO, 3/5/9 GND, 4 SWCLK, 6 SWO, 7/8 not used, 10 NRST




USART2: 1 TX (PA2), 2 RX (PA3), 3 GND




Test points: SPI1 (LAN9252), SPI2 (DAC), relay OE, analog clamp, MPG A/B and the spare pin PH1







STM32F405VGT6 (LQFP-100), shared clock, SWD, UART, heartbeat LED

**Yantra Innovation — LCEC-IO**

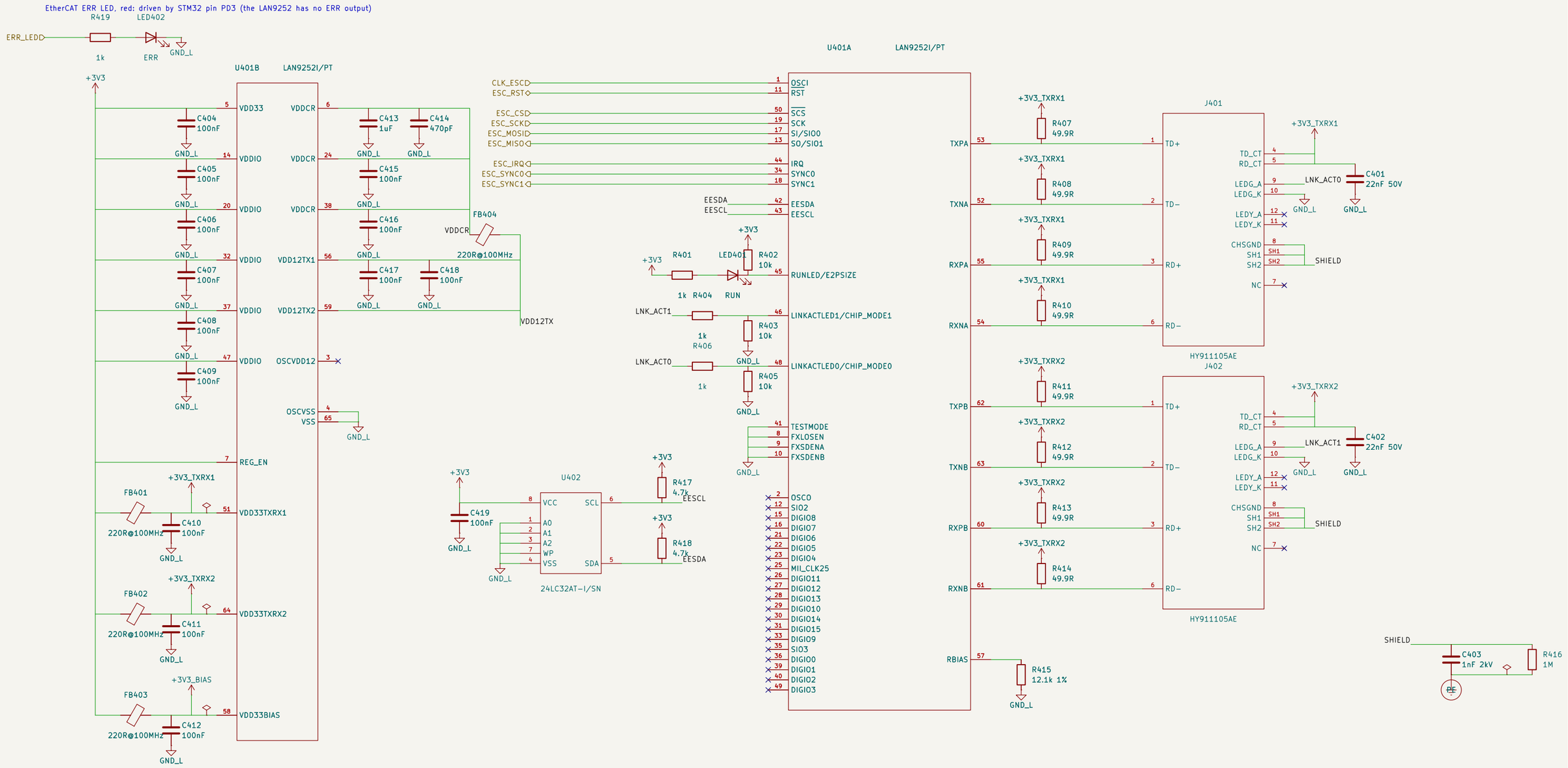
Sheet: /MCU/
File: 03_mcu.kicad_sch

***Title: LCEC-IO — 3 MCU***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 3/13 |

# 4 ETHERCAT — LAN9252 (SPI), ESI EEPROM, two RJ45 jacks with magnetics, RUN and ERR LEDs

Chip mode 00 = 2 ports (PHY A = port 0 = J401, PHY B = port 1 = J402). The PDI is chosen by the ESI EEPROM content (SPI slave). Rows and values follow the LAN9252 datasheet DS00001909A (sections 3, 4, 7, 12.10 and 14).




**Design notes**

RUNLED / E2PSIZE: 10 k pull-up = EEPROM size strap high (32 Kbit and up, 2-byte addressing). The pin is open drain, so the RUN LED is +3V3 > 1 k > LED > pin.

LINKACTLED0 / CHIP_MODE0 and LINKACTLED1 / CHIP_MODE1: 10 k pull-downs = chip mode 00 (2 ports) and active-high LED outputs; 1 k to the green LED in each jack.

TESTMODE to ground (required). FXLOSEN, FXSDENA and FXSDENB to ground = copper twisted pair. REG_EN to 3.3 V = internal 1.2 V regulators. OSCO open (single-ended oscillator on OSCI).

Pins 12, 15 to 16, 21 to 23, 25 to 31, 33, 35, 36, 39, 40 and 49 (host bus, MII, digital I/O) are unused in SPI mode and left open.

As in the EVB-LAN9252-SPI reference: VDD33TXRX1, VDD33TXRX2 and VDD33BIAS each come from +3V3 through a 220 R ferrite bead with 100 nF; VDD12TX1/2 come from VDDCR through a ferrite bead with 2 x 100 nF.

Each twisted-pair line has 49.9 R to its port's filtered 3.3 V; the magnetics centre taps (TXCT and RXCT together) are on the same net with 22 nF. RBIAS 12.1 k 1 % to ground. VDDCR: 1 uF + 470 pF at pin 6, 100 nF at pins 24 and 38.

The ESI EEPROM must hold PDI = SPI before the first run. WP is low so the master can write it (the EVB ties WP high).

LAN9252 in SPI mode, 24LC32A ESI EEPROM, two RJ45 jacks, RUN and ERR LEDs

**Yantra Innovation — LCEC-IO**

Sheet: /EtherCAT/

File: 04_ethercat.kicad_sch

***Title: LCEC-IO — 4 EtherCAT***

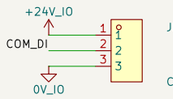
| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 4/13 |

# 5 INPUTS – 8 digital inputs (DI) and 12 MPG selector inputs (SEL), 24 VDC, PNP or NPN

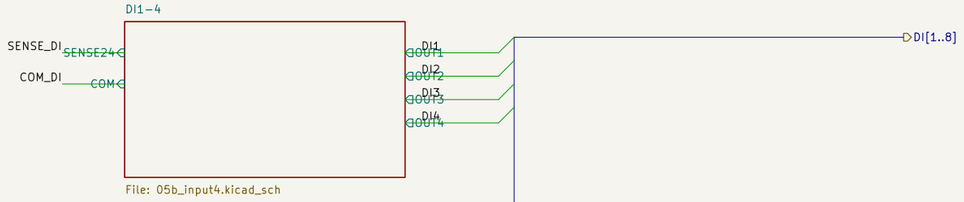
Five copies of the 4-channel sheet. Per group: sensor +24 V through a 0.5 A PTC and one COM jumper (1–2: COM to +24V_IO for NPN sensors, 2–3: COM to 0V_IO for PNP sensors, the default). Dry-contact switches wire between +24 V and the input.

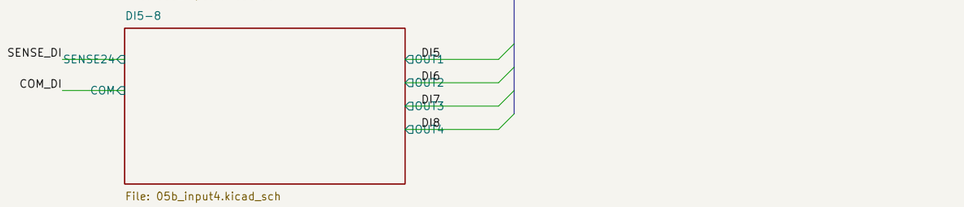
## DI group sensor supply



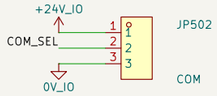

JS501 jumper shunt on JP501 (2–3 = PNP)



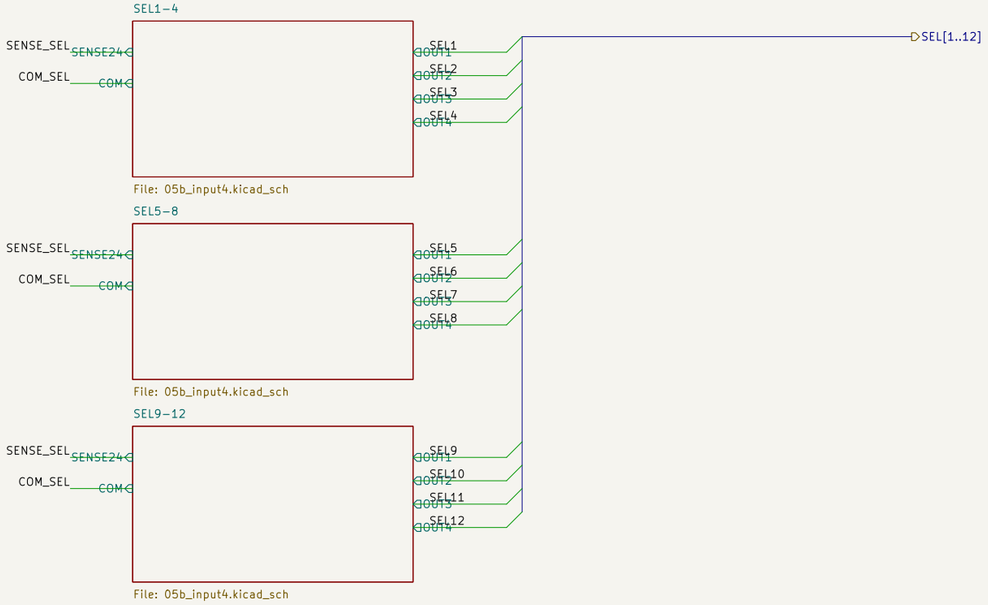




## SEL group sensor supply




JS502 jumper shunt on JP502 (2–3 = PNP)




## Accessories, not on the PCB: 3-pole plugs for J501 to J520

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| P501 | 3-pole plug for J501 | P506 | 3-pole plug for J506 | P511 | 3-pole plug for J511 | P516 | 3-pole plug for J516 |
| P502 | 3-pole plug for J502 | P507 | 3-pole plug for J507 | P512 | 3-pole plug for J512 | P517 | 3-pole plug for J517 |
| P503 | 3-pole plug for J503 | P508 | 3-pole plug for J508 | P513 | 3-pole plug for J513 | P518 | 3-pole plug for J518 |
| P504 | 3-pole plug for J504 | P509 | 3-pole plug for J509 | P514 | 3-pole plug for J514 | P519 | 3-pole plug for J519 |
| P505 | 3-pole plug for J505 | P510 | 3-pole plug for J510 | P515 | 3-pole plug for J515 | P520 | 3-pole plug for J520 |

8 digital inputs and 12 selector inputs: sensor supply PTCs, COM jumpers, five 4-channel sheets

**Yantra Innovation – LCEC-IO**

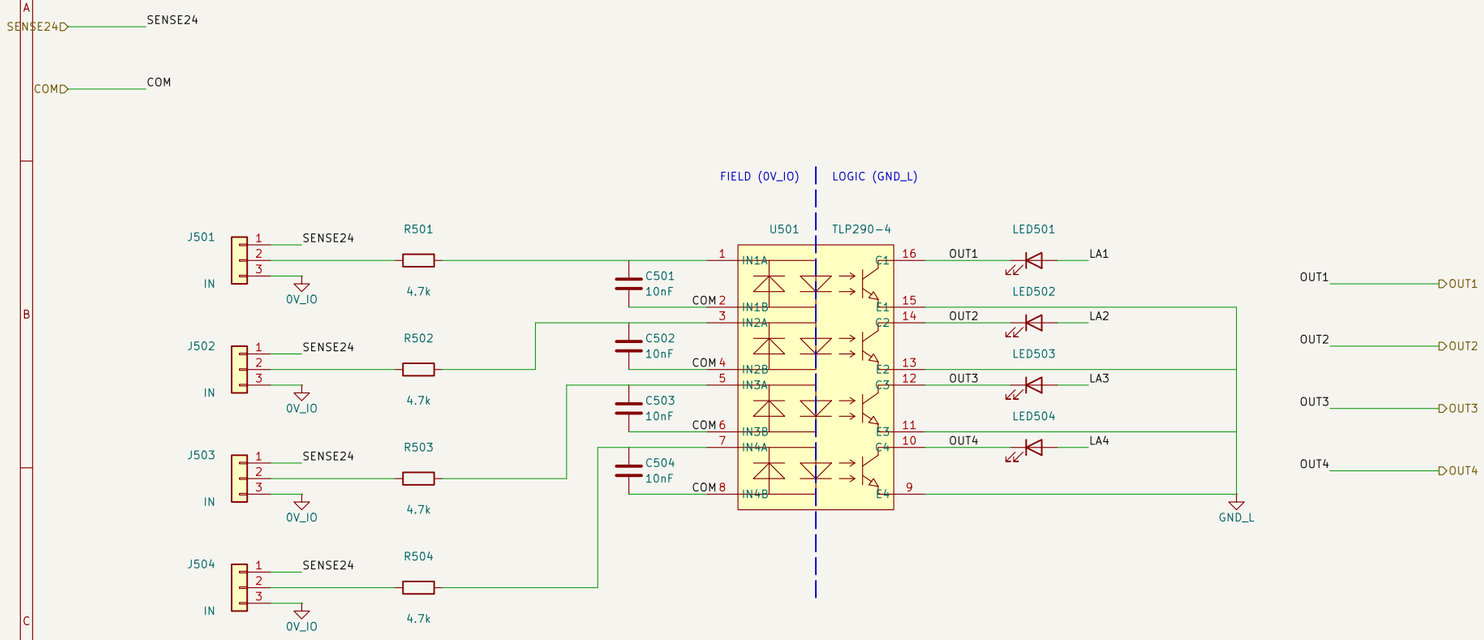
Sheet: /Inputs/
File: 05_inputs.kicad_sch

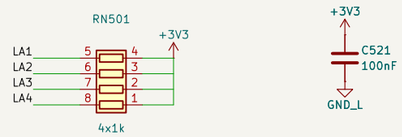
***Title: LCEC-IO – 5 Inputs***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-11-06 | **Rev: A** |
| KiCad E.D.A. 10.0.8 | | Id: 5/13 |

# 5 INPUTS – 4 channels (one TLP290-4): DI1-4, DI5-8, SEL1-4, SEL5-8 and SEL9-12 use this sheet

AC-input optocoupler: one COM jumper per group (parent sheet) selects PNP (COM = 0V_IO) or NPN (COM = +24V_IO) sensors. Per channel 4.7 k in series and 10 nF across the optocoupler input (47 us filter).




4 x 24 V input: TLP290-4 AC-input optocoupler, 4.7 k + 10 nF filter, state LED; placed 5 times

**Yantra Innovation – LCEC-IO**

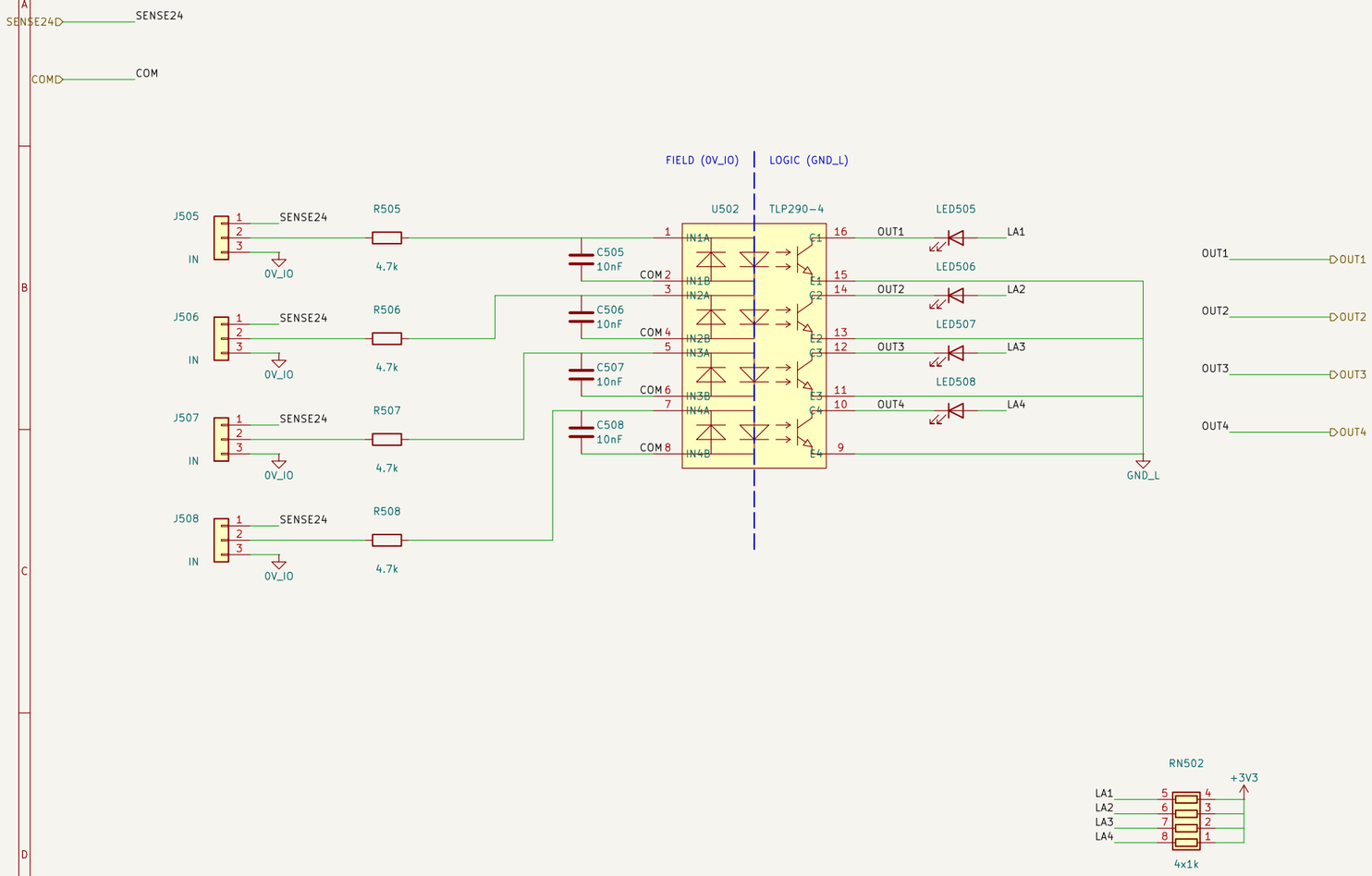
Sheet: /Inputs/DI1-4/
File: 05b_input4.kicad_sch

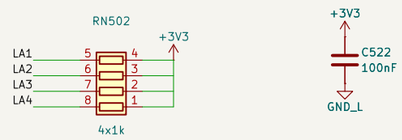
***Title: LCEC-IO – 5 Inputs (4 channels)***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 5.1/13 |

# 5 INPUTS – 4 channels (one TLP290-4): DI1-4, DI5-8, SEL1-4, SEL5-8 and SEL9-12 use this sheet

AC-input optocoupler: one COM jumper per group (parent sheet) selects PNP (COM = 0V_IO) or NPN (COM = +24V_IO) sensors. Per channel 4.7 k in series and 10 nF across the optocoupler input (47 us filter).




4 x 24 V input: TLP290-4 AC-input optocoupler, 4.7 k + 10 nF filter, state LED; placed 5 times

**Yantra Innovation – LCEC-IO**

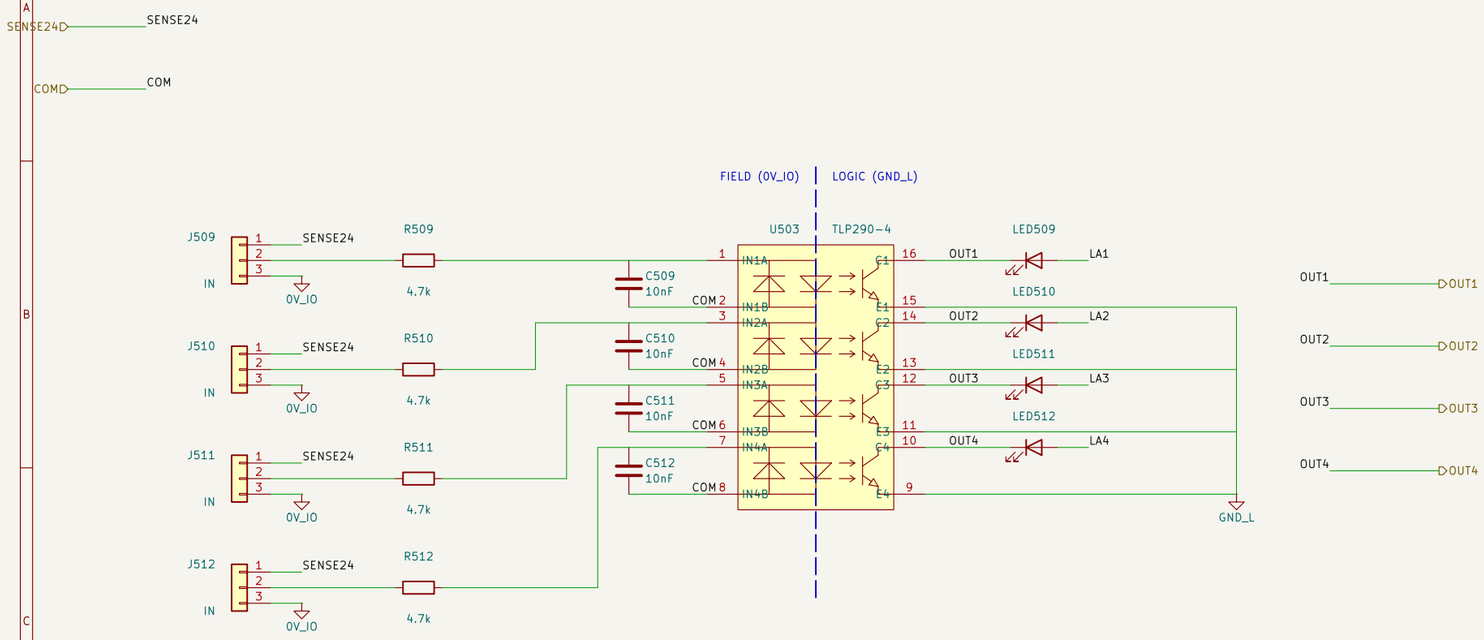
Sheet: /Inputs/DI5-8/
File: 05b_input4.kicad_sch

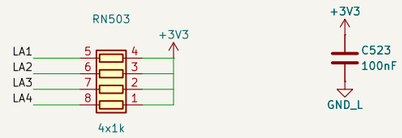
***Title: LCEC-IO – 5 Inputs (4 channels)***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 5.2/13 |

# 5 INPUTS – 4 channels (one TLP290-4): DI1-4, DI5-8, SEL1-4, SEL5-8 and SEL9-12 use this sheet

AC-input optocoupler: one COM jumper per group (parent sheet) selects PNP (COM = 0V_IO) or NPN (COM = +24V_IO) sensors. Per channel 4.7 k in series and 10 nF across the optocoupler input (47 us filter).




4 x 24 V input: TLP290-4 AC-input optocoupler, 4.7 k + 10 nF filter, state LED; placed 5 times

**Yantra Innovation – LCEC-IO**

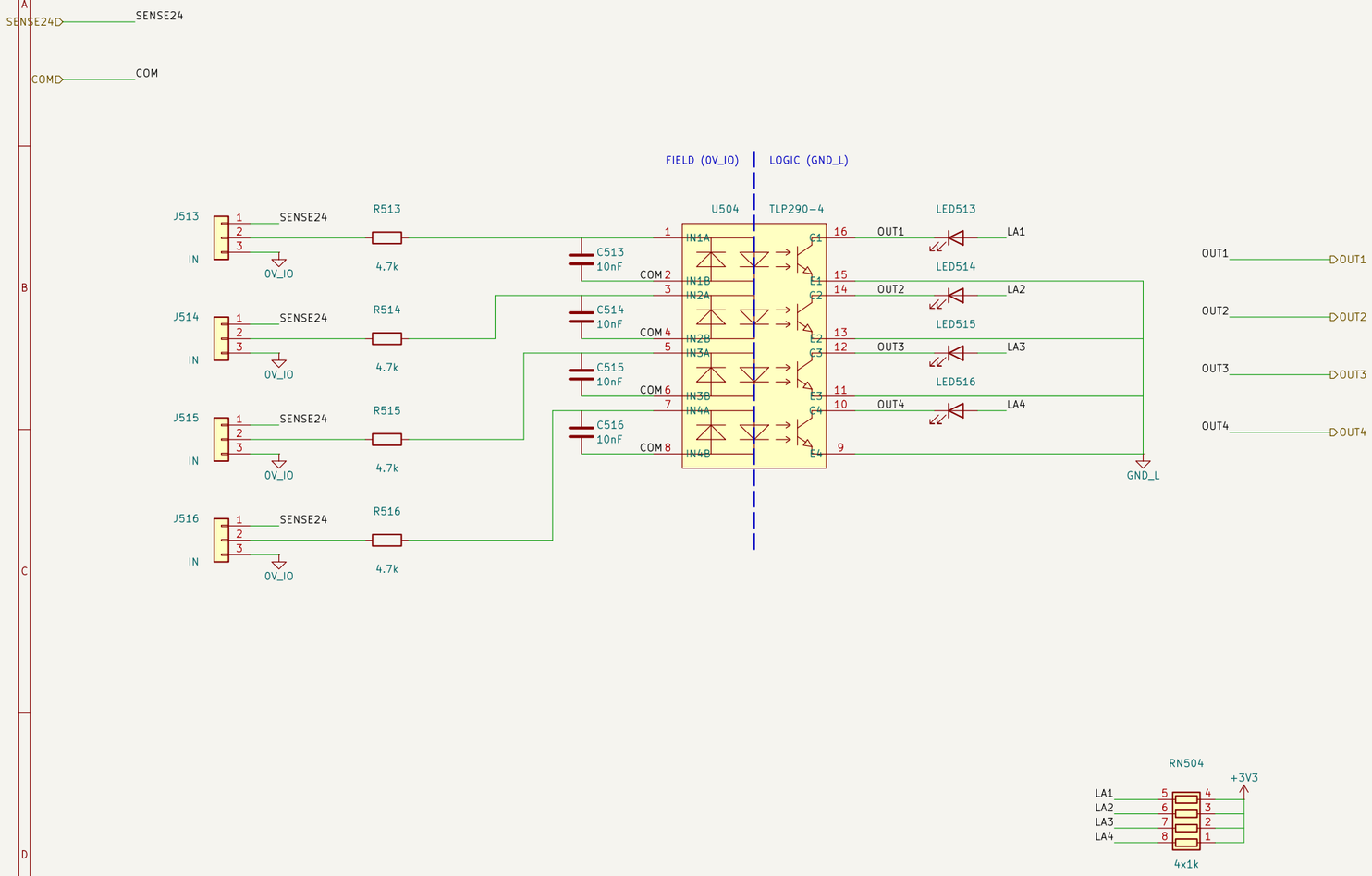
Sheet: /Inputs/SEL1-4/
File: 05b_input4.kicad_sch

***Title: LCEC-IO – 5 Inputs (4 channels)***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 5.3/13 |

# 5 INPUTS – 4 channels (one TLP290-4): DI1-4, DI5-8, SEL1-4, SEL5-8 and SEL9-12 use this sheet

AC-input optocoupler: one COM jumper per group (parent sheet) selects PNP (COM = 0V_IO) or NPN (COM = +24V_IO) sensors. Per channel 4.7 k in series and 10 nF across the optocoupler input (47 us filter).




4 x 24 V input: TLP290-4 AC-input optocoupler, 4.7 k + 10 nF filter, state LED; placed 5 times

**Yantra Innovation – LCEC-IO**

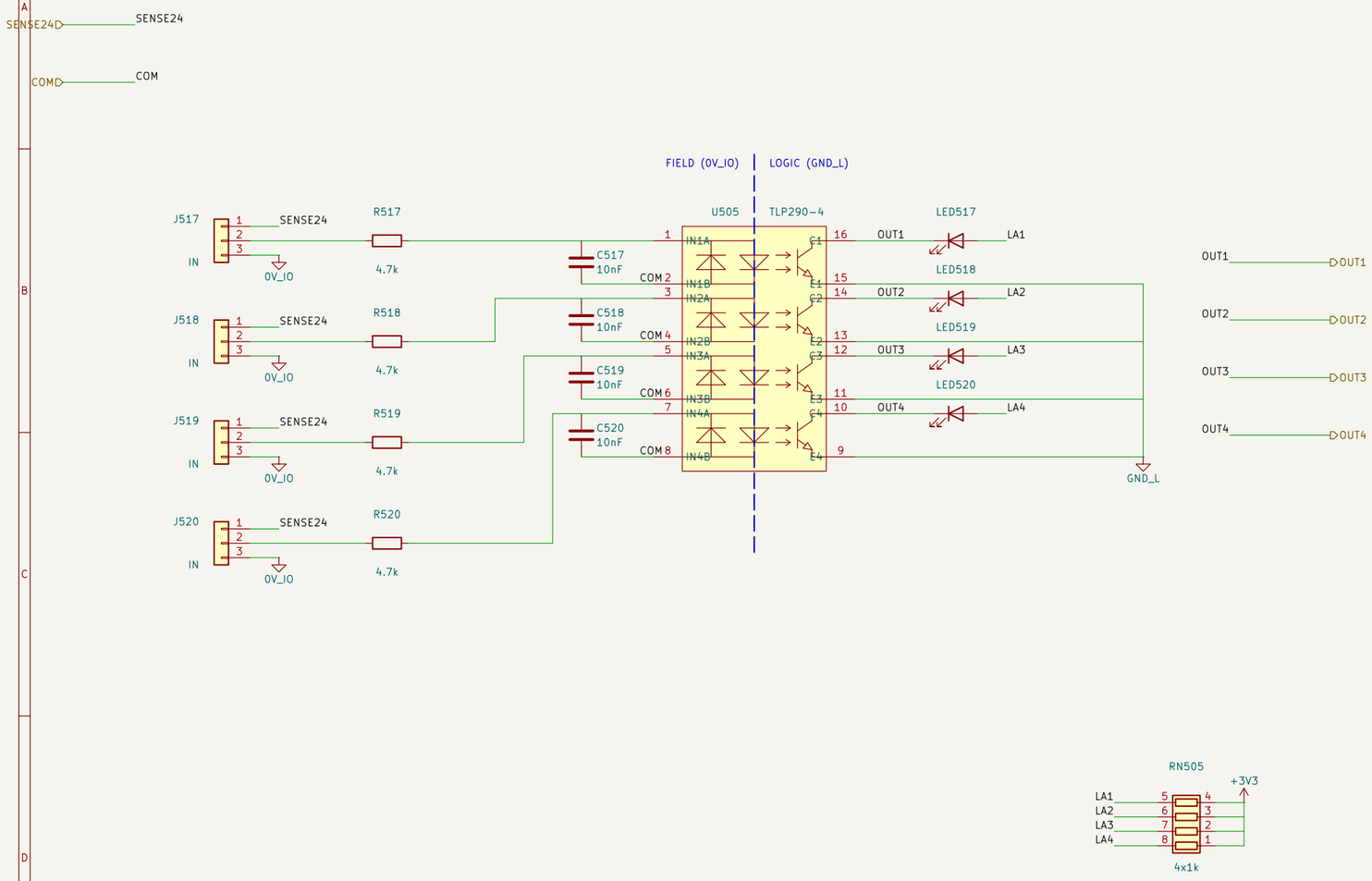
Sheet: /Inputs/SEL5-8/
File: 05b_input4.kicad_sch

***Title: LCEC-IO – 5 Inputs (4 channels)***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 5.4/13 |

# 5 INPUTS – 4 channels (one TLP290-4): DI1-4, DI5-8, SEL1-4, SEL5-8 and SEL9-12 use this sheet

AC-input optocoupler: one COM jumper per group (parent sheet) selects PNP (COM = 0V_IO) or NPN (COM = +24V_IO) sensors. Per channel 4.7 k in series and 10 nF across the optocoupler input (47 us filter).




4 x 24 V input: TLP290-4 AC-input optocoupler, 4.7 k + 10 nF filter, state LED; placed 5 times

**Yantra Innovation – LCEC-IO**

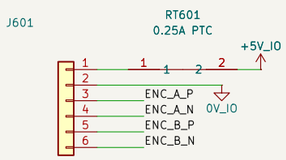
Sheet: /Inputs/SEL9-12/
File: 05b_input4.kicad_sch

***Title: LCEC-IO – 5 Inputs (4 channels)***

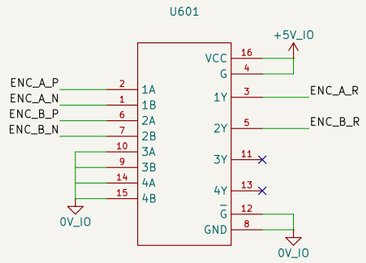
| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 5.5/13 |

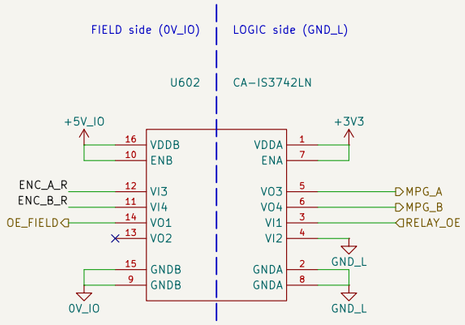
# 6 MPG — 5 V differential pulse generator input, isolated

A / A-bar / B / B-bar from the pendant are received by an AM26LS32A, isolated by a CA-IS3742LN and sent to PB4 / PB5 (TIM3 encoder). The same isolator carries the relay output enable (forward channel 1).

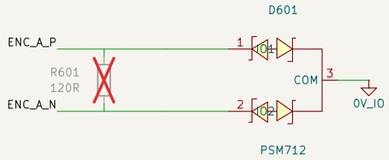



1 +5V_IO  2 0V_IO  3 A  4 A-bar  5 B  6 B-bar




TVS pairs and unfitted 120 R terminations (20 kHz over 5 m does not need them)



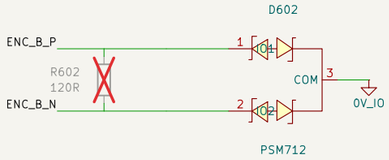













P601  6-pole cable plug for J601

**Design notes**

AM26LS32A (0 to 70 C, SOIC-16) supplied from +5V_IO; G high and ~G low enable all receivers; the unused receivers 3 and 4 have their inputs grounded.

RT601 limits the pendant supply to 0.25 A hold. PSM712 clamps each line pair (−7 V / +12 V). The 120 R terminations are not fitted.

CA-IS3742LN: reverse channels 3 and 4 carry A and B to PB4 / PB5 (default output low). Forward channel 1 carries the relay output enable; channel 2 is spare, input tied low.

MPG 5 V differential receiver, isolated to the MCU; also carries the relay output enable

**Yantra Innovation — LCEC-IO**

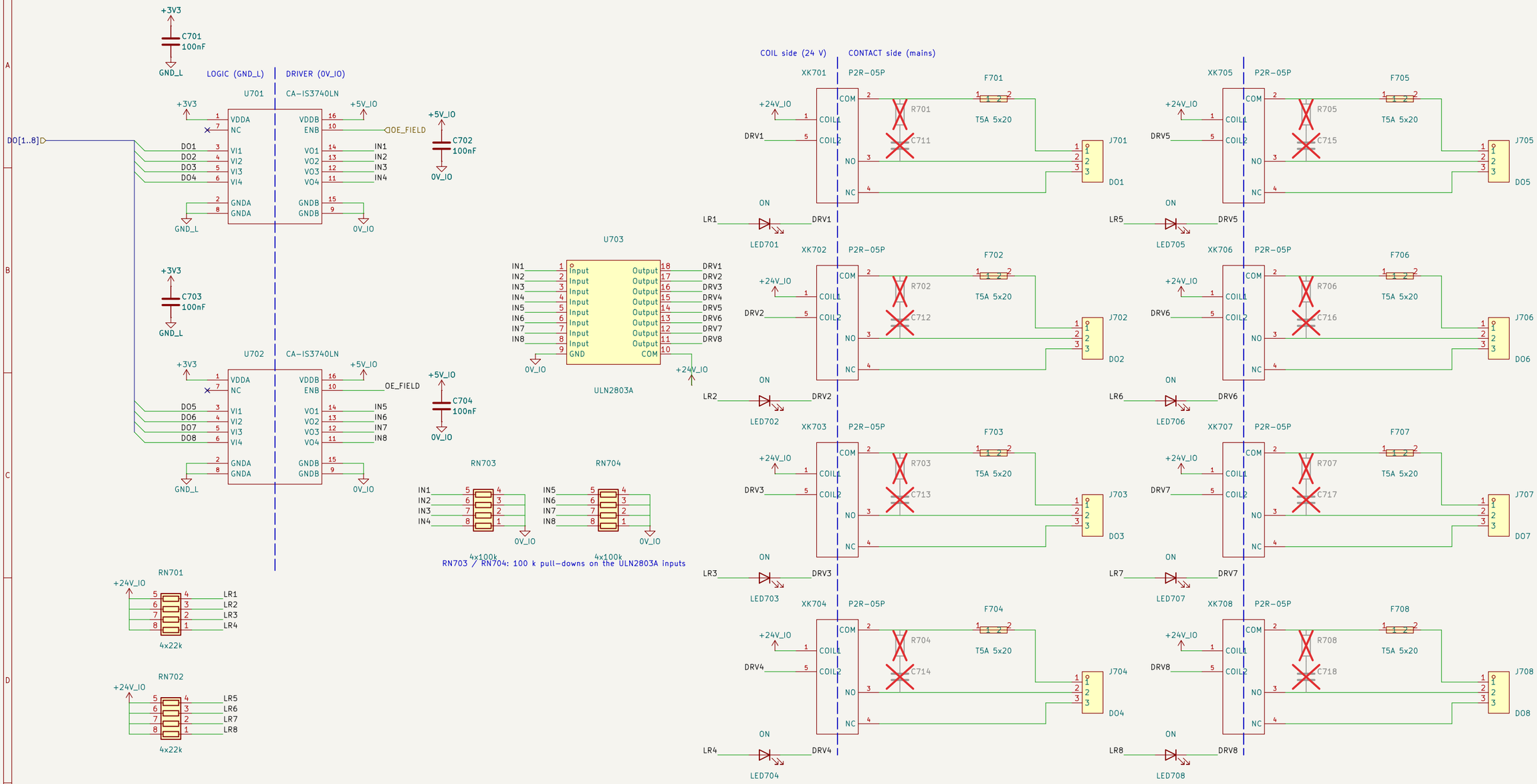
Sheet: /MPG/
File: 06_mpg.kicad_sch

***Title: LCEC-IO — 6 MPG***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 6/13 |

# 7 RELAY OUTPUTS — 8 x Omron G2R-1-S DC24 in P2R-05P sockets, fuse and ON LED per channel

DO1 to DO8 pass two CA-IS3740LN isolators (output enable OE_FIELD from the MPG sheet) to one ULN2803A. All outputs are off when OE is low, the MCU resets or the logic supply is lost.




**Accessories, not on the PCB (in the BOM)**

| | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| K701 | G2R-1-S DC24 relay (local) | FU701 | T5A 5x20 ceramic fuse (local) | FC701 | clear cover for F701 | P701 | 3-pole plug for J701 |
| K702 | G2R-1-S DC24 relay (local) | FU702 | T5A 5x20 ceramic fuse (local) | FC702 | clear cover for F702 | P702 | 3-pole plug for J702 |
| K703 | G2R-1-S DC24 relay (local) | FU703 | T5A 5x20 ceramic fuse (local) | FC703 | clear cover for F703 | P703 | 3-pole plug for J703 |
| K704 | G2R-1-S DC24 relay (local) | FU704 | T5A 5x20 ceramic fuse (local) | FC704 | clear cover for F704 | P704 | 3-pole plug for J704 |
| K705 | G2R-1-S DC24 relay (local) | FU705 | T5A 5x20 ceramic fuse (local) | FC705 | clear cover for F705 | P705 | 3-pole plug for J705 |
| K706 | G2R-1-S DC24 relay (local) | FU706 | T5A 5x20 ceramic fuse (local) | FC706 | clear cover for F706 | P706 | 3-pole plug for J706 |
| K707 | G2R-1-S DC24 relay (local) | FU707 | T5A 5x20 ceramic fuse (local) | FC707 | clear cover for F707 | P707 | 3-pole plug for J707 |
| K708 | G2R-1-S DC24 relay (local) | FU708 | T5A 5x20 ceramic fuse (local) | FC708 | clear cover for F708 | P708 | 3-pole plug for J708 |

8 relay outputs: isolated drive, ULN2803A, plug-in relays, fuse and ON LED per channel

**Yantra Innovation – LCEC-IO**

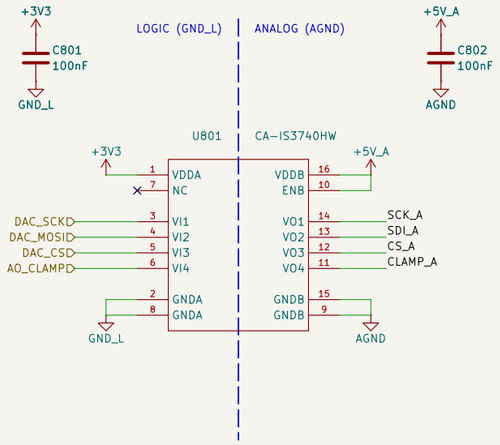
Sheet: /Relays/
File: 07_relays.kicad_sch

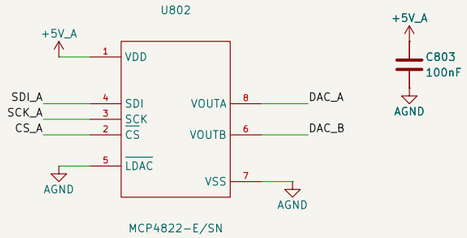
***Title: LCEC-IO – 7 Relays***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-11-06 | **Rev: A** |
| KiCad E.D.A. 10.0.8 | | Id: 7/13 |

# 8 ANALOG OUTPUTS — 2 x 0 to 10 V, 12 bit, isolated, 0 V at power-up and on loss

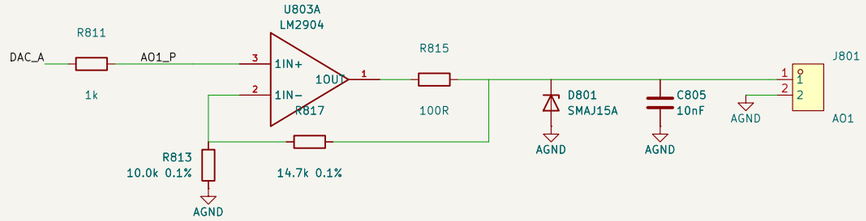
SPI and the clamp signal pass a CA-IS3740HW (default high: DAC deselected, outputs clamped) to an MCP4822 on +5V_A (78L05 from +15V_A). A clamp MOSFET holds both op-amp inputs at AGND until firmware releases AO_CLAMP.

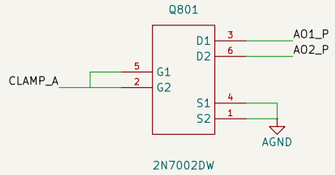


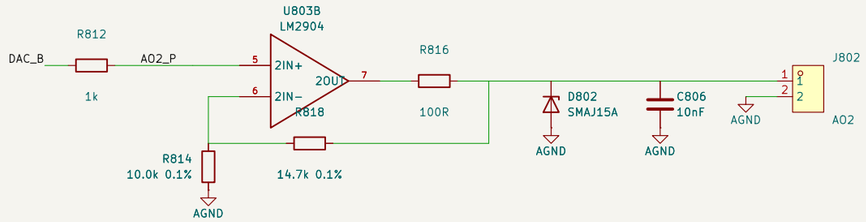



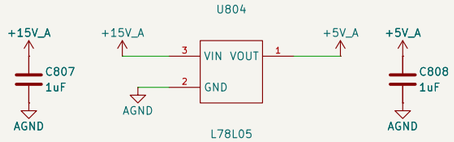


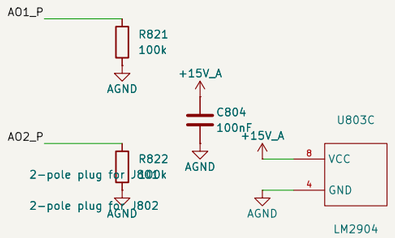





P801 2-pole plug for J801

P802 2-pole plug for J802

**Design notes**

Gain 1 + 14.7 k / 10.0 k = 2.47 (0.1 % resistors): DAC full scale 4.095 V gives 10.11 V; firmware calibrates the 0 V and 10 V points of each channel.

Output 100 R inside the feedback loop, SMAJ15A and 10 nF to AGND. At 10 V into 2 k the op-amp swings to 10.6 V, below its 13.5 V limit on 15 V.

Accuracy after calibration: reference drift 125 ppm/C x 55 C = 0.7 %, offset 3 mV = 0.03 %, resistors 0.2 %: about +/-0.9 % FS worst case over 0 to 80 C.

CA-IS3740HW outputs default high: CS_A high deselects the DAC and CLAMP_A high turns the clamp MOSFET on, so both outputs sit at 0 V at power-up and when logic power is lost.

2 x 0-10 V outputs: isolated SPI, MCP4822 DAC, LM2904 gain stage, power-up clamp

**Yantra Innovation — LCEC-IO**

Sheet: /Analog/
File: 08_analog.kicad_sch

***Title: LCEC-IO — 8 Analog***

| Size: A3 | Date: 2026-11-06 | **Rev: A** |
|---|---|---|
| KiCad E.D.A. 10.0.8 | | Id: 8/13 |